

Red Hat is part of the HPE GreenLake system now

Hewlett Packard Enterprise has announced that Red Hat, a long-time partner, has joined the HPE GreenLake ecosystem. The companies will collaborate to combine new service offerings with open source technologies from Red Hat, such as Red Hat OpenShift, Red Hat Enterprise Linux, and Red Hat Ansible Automation Platform, all of which will be deployed on the HPE GreenLake platform.

The combination of the HPE GreenLake edge-to-cloud platform and Red Hat's enterprise open source technologies will assist IT organisations in meeting their need for agile technology solutions that can accelerate business outcomes and fuel their digital transformations. HPE and Red Hat will collaborate to deliver enterprise-ready, open source solutions that reduce costs and complexity while improving stability and performance across multiple IT environments.

The partnership will provide a number of advantages. Platforms and solutions designed for a variety of workloads will provide more reliable infrastructure with enhanced security, regardless of location. They will have a lower risk of complexity and lower cost of ownership.



Businesses that use these complete, open source IT stack solutions can move faster and more quickly adapt to market and industry changes, resulting in a shorter time to market and a competitive advantage through on-demand capabilities.

Stanford University open sources AI based Diffusion-LM

Diffusion-LM, a non-autoregressive generative language model that allows for fine-grained control of the model's output text, has been open sourced by Stanford University researchers. This model supposedly outperforms existing methods on controlled text generation tasks.

The model and experiments were described in an arXiv paper. Diffusion-LM is a generative language model that employs a plug-and-play control scheme in which the language model is fixed and the generated text is guided by an external classifier that determines how well the generated text matches the desired parameters. Users can specify the required parts of speech, syntax tree, and sentence length for the desired output.



Many autoregressive generative language models (LM), such as GPT-3, recursively generate text by predicting the next word in a sequence, then adding that word to the existing sequence and using the updated sequence as input for further prediction. These models can generate text that is indistinguishable from human-written text, and they can generate text to solve a variety of problems ranging from question-answering to interactive chat. However, providing any user control over the generated output, such as a desired sentence length, structure, or sentiment, is difficult.

One possible solution is to fine-tune the LM to accept an additional control input, but this update can be computationally intensive and may not generalise to handle multiple control parameters. Another approach is to use a plug-and-play technique that freezes the LM's parameters and steers the generation with an external classifier that evaluates how close the generated output is to the desired parameters. Attempts to steer autoregressive models, on the other hand, have proven difficult.

Instead of attempting to steer an autoregressive LM, the Stanford researchers chose a new language generation technique: a diffusion model. These models have performed well in computer vision and other continuous domains, but have not been applied to text generation, which is a discrete domain. The team claims that Diffusion-LM is the first diffusion model for text generation.

Diffusion-LM was tested on five classifier-guided text generation control tasks and compared to baseline methods using a GPT-2 autoregressive LM, with both plug-and-play and fine-tuning.

DigiKam releases version 7.7

DigiKam, an open source multi-platform digital photo management application, has released version 7.7 for Windows, macOS, and Linux, which includes new features,